

Exercise 5

Ziyu Ma

Part1:

Task1:

Question1: What is the advantage of representing rigid transformations with dual quaternions for blending?

Answer: It offers advantages due to their ability to handle translations and rotations simultaneously, and it also more unified and efficient e.g. in normalization.

Task2:

Question3:

To prove $\hat{\mathbf{q}}^t = \cos(t \frac{\hat{\theta}}{2}) + \hat{\mathbf{s}} \sin(t \frac{\hat{\theta}}{2})$, we first use the formula $\hat{\mathbf{q}}^t = e^{t \log(\hat{\mathbf{q}})}$. $\hat{\mathbf{q}} = \cos(\frac{\hat{\theta}}{2}) + \hat{\mathbf{s}} \sin(\frac{\hat{\theta}}{2})$ and $\log(\cos(\frac{\hat{\theta}}{2}) + \hat{\mathbf{s}} \sin(\frac{\hat{\theta}}{2})) = \frac{\hat{\theta}}{2}$. We get $\hat{\mathbf{q}}^t = e^{t \frac{\hat{\theta}}{2}}$.

By knowing $\hat{\theta} = \theta_0 + \epsilon \theta_\epsilon$ and $\hat{\mathbf{s}} = \mathbf{s}_0 + \epsilon \mathbf{s}_\epsilon$, we know:

$$\begin{aligned} \hat{\mathbf{s}} \hat{\theta} &= (\theta_0 + \epsilon \theta_\epsilon)(\mathbf{s}_0 + \epsilon \mathbf{s}_\epsilon) \\ &= \theta_0 \mathbf{s}_0 + \epsilon \theta_0 \mathbf{s}_\epsilon + \epsilon \theta_\epsilon \mathbf{s}_0 \\ &= \theta_0 \mathbf{s}_0 + \epsilon(\theta_0 \mathbf{s}_\epsilon + \theta_\epsilon \mathbf{s}_0) \end{aligned}$$

The term with ϵ^2 vanishes, because $\epsilon^2 = 0$.

Using $e^{\hat{\mathbf{q}}} = \cos(\|\hat{\mathbf{q}}\|) + \frac{\hat{\mathbf{q}}}{\|\hat{\mathbf{q}}\|} \sin(\|\hat{\mathbf{q}}\|)$, we get:

$$e^{t \frac{\hat{\theta}}{2}} = \cos(\|t \frac{\hat{\theta}}{2}\|) + \frac{t \frac{\hat{\theta}}{2}}{\|t \frac{\hat{\theta}}{2}\|} \sin(\|t \frac{\hat{\theta}}{2}\|)$$

So we need to get $\|t \frac{\hat{\theta}}{2}\|$. The formula $\|\hat{\mathbf{q}}\| = \|\mathbf{q}_0\| + \epsilon \frac{\langle \mathbf{q}_0, \mathbf{q}_\epsilon \rangle}{\|\mathbf{q}_0\|}$ is useful for that.

$$\|t \frac{\hat{\theta}}{2}\| = \|\frac{t}{2} \theta_0 \mathbf{s}_0 + \epsilon \frac{t}{2} (\theta_0 \mathbf{s}_\epsilon + \theta_\epsilon \mathbf{s}_0)\|$$

, so $\mathbf{q}_0$ is $\frac{t}{2} \theta_0 \mathbf{s}_0$ and $\mathbf{q}_\epsilon$ is $\frac{t}{2} (\theta_0 \mathbf{s}_\epsilon + \theta_\epsilon \mathbf{s}_0)$. Thus we get:

$$\begin{aligned} &= \|\frac{t}{2} \theta_0 \mathbf{s}_0\| + \epsilon \frac{\langle \frac{t}{2} \theta_0 \mathbf{s}_0, \frac{t}{2} (\theta_0 \mathbf{s}_\epsilon + \theta_\epsilon \mathbf{s}_0) \rangle}{\|\frac{t}{2} \theta_0 \mathbf{s}_0\|} \\ &= \|\frac{t}{2} \theta_0 \mathbf{s}_0\| + \epsilon \frac{\langle \frac{t}{2} \theta_0 \mathbf{s}_0, \frac{t}{2} \theta_0 \mathbf{s}_\epsilon \rangle + \langle \frac{t}{2} \theta_0 \mathbf{s}_0, \frac{t}{2} \theta_\epsilon \mathbf{s}_0 \rangle}{\|\frac{t}{2} \theta_0 \mathbf{s}_0\|} \end{aligned}$$

By using $\langle \mathbf{s}_0, \mathbf{s}_\epsilon \rangle = 0$ and $\langle \mathbf{s}_0, \mathbf{s}_0 \rangle = 1$, we get:

$$\begin{aligned}
&= \left\| \frac{t}{2} \theta_0 \mathbf{s}_0 \right\| + \epsilon \frac{\langle \frac{t}{2} \theta_0 \mathbf{s}_0, \frac{t}{2} \theta_\epsilon \mathbf{s}_0 \rangle}{\left\| \frac{t}{2} \theta_0 \mathbf{s}_0 \right\|} \\
&= \frac{t}{2} \theta_0 + \epsilon \frac{\frac{t^2}{4} \theta_0 \theta_\epsilon}{\frac{t}{2} \theta_0} \\
&= \frac{t}{2} \theta_0 + \epsilon \frac{t}{2} \theta_\epsilon \\
&= \frac{t}{2} (\theta_0 + \epsilon \theta_\epsilon) \\
&= \frac{t}{2} \hat{\theta}
\end{aligned}$$

Thus the following formula becomes:

$$\begin{aligned}
e^{t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}} &= \cos(\|t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}\|) + \frac{t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}}{\|t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}\|} \sin(\|t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}\|) \\
&= \cos\left(\frac{t}{2} \hat{\theta}\right) + \frac{t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}}{\frac{t \hat{\theta}}{2}} \sin\left(\frac{t}{2} \hat{\theta}\right) \\
&= \cos\left(\frac{t}{2} \hat{\theta}\right) + \hat{\mathbf{s}} \sin\left(\frac{t}{2} \hat{\theta}\right)
\end{aligned}$$

Since $\hat{\mathbf{q}}^t = e^{t \hat{\mathbf{s}} \frac{\hat{\theta}}{2}}$, we get:

$$\hat{\mathbf{q}}^t = \cos\left(\frac{t}{2} \hat{\theta}\right) + \hat{\mathbf{s}} \sin\left(\frac{t}{2} \hat{\theta}\right)$$

And we proved the required equation!

Question4:

Dual quaternion $\hat{\mathbf{q}} = \mathbf{q}_0 + \epsilon \mathbf{q}_\epsilon$ that rotates by $\mathbf{q}_0$ and translates by $\mathbf{t}$

$$\hat{\mathbf{q}} = \left(1 + \frac{\epsilon}{2} (t_0 \mathbf{i} + t_1 \mathbf{j} + t_2 \mathbf{k}) \right) \mathbf{q}_0$$

$$\hat{\mathbf{q}} = \mathbf{q}_0 + \frac{\epsilon}{2} (t_0 \mathbf{i} + t_1 \mathbf{j} + t_2 \mathbf{k}) \mathbf{q}_0$$

is always a unit dual quaternion.

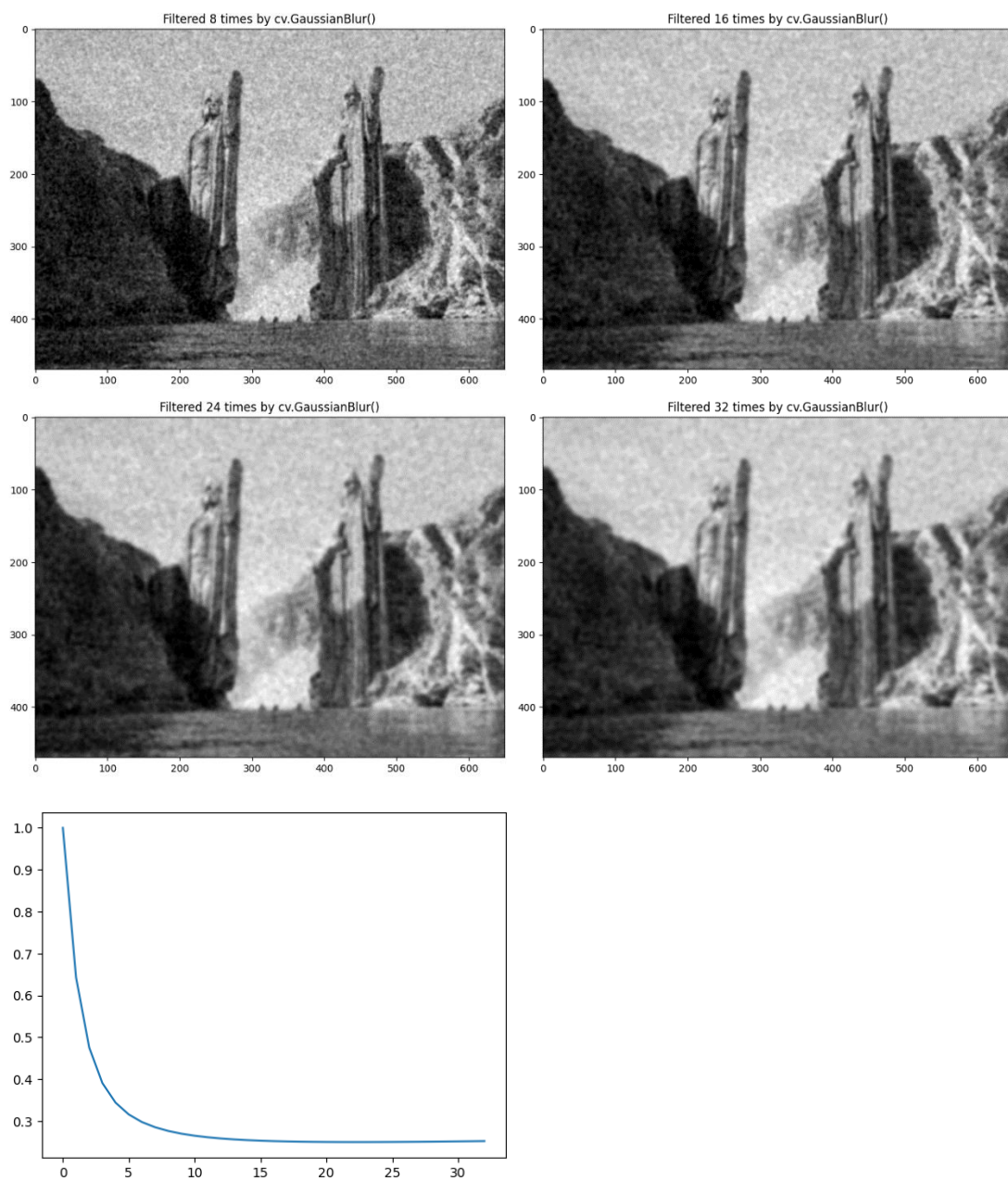
As the formula in the lecture slide, rotation and translation are separate. The axis to rotate (z-axis) and angle to rotate is included in $\mathbf{q}_0$ while what to translate is included in $\mathbf{t}$, so we can just translate the object from the second to the third image using the $\mathbf{t}$ vector.

Part2:

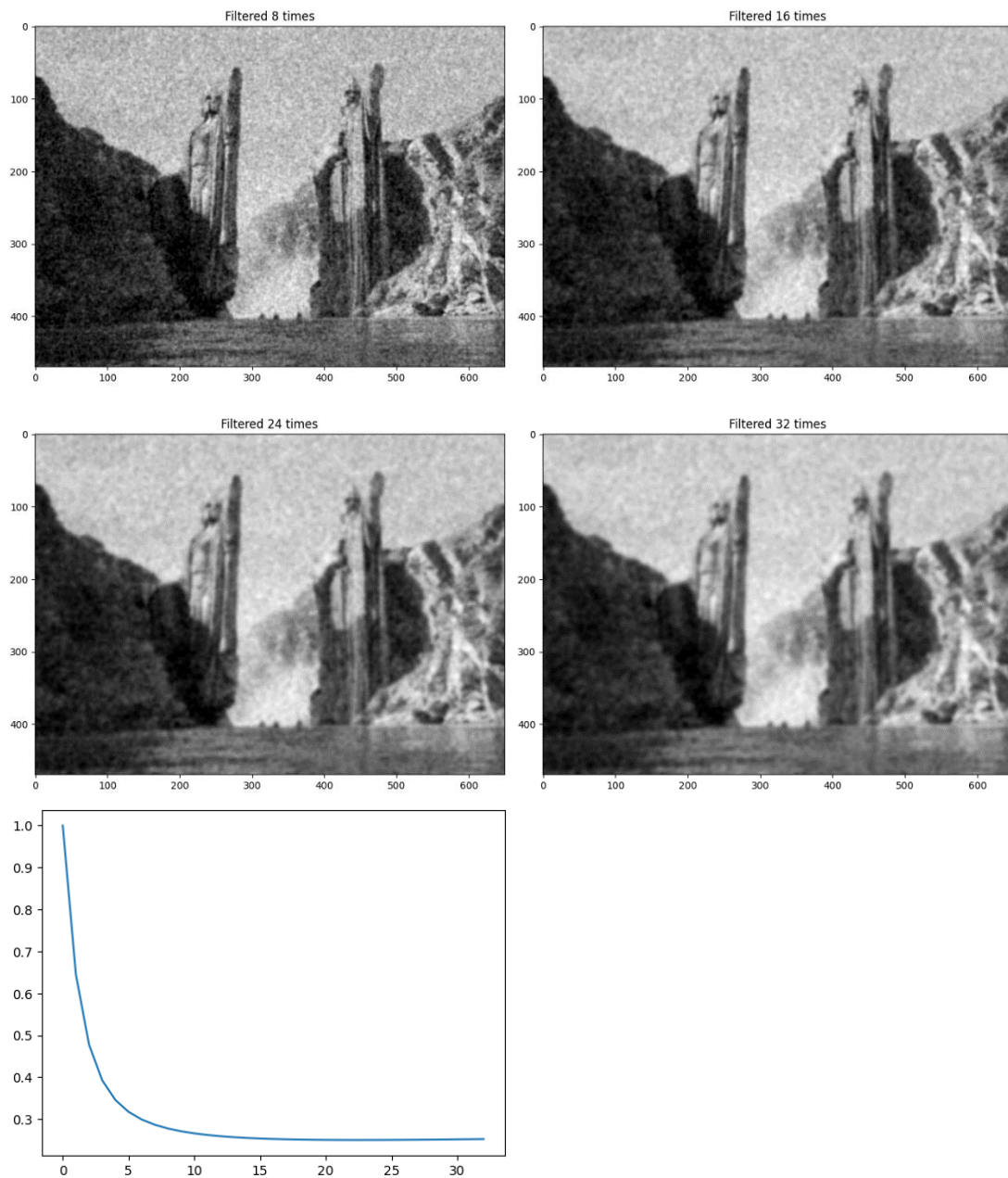
For this part, I used Jupyter notebook because I met some problem with installing opencv on my machine. With conda list, opencv is in the list, however "import cv2" cannot be identified. So I used Google Colab to run my code. Please also use Google Colab to test my code, and change the path of the picture. Thanks!

Task1: Filtering

I use `cv2.GaussianBlur()` with standard deviation = 0.5, size = 5 and borderType = `cv2.BORDER_REFLECT_101` to simulate Neumann boundary conditions (where the gradient at the border of the image is zero).



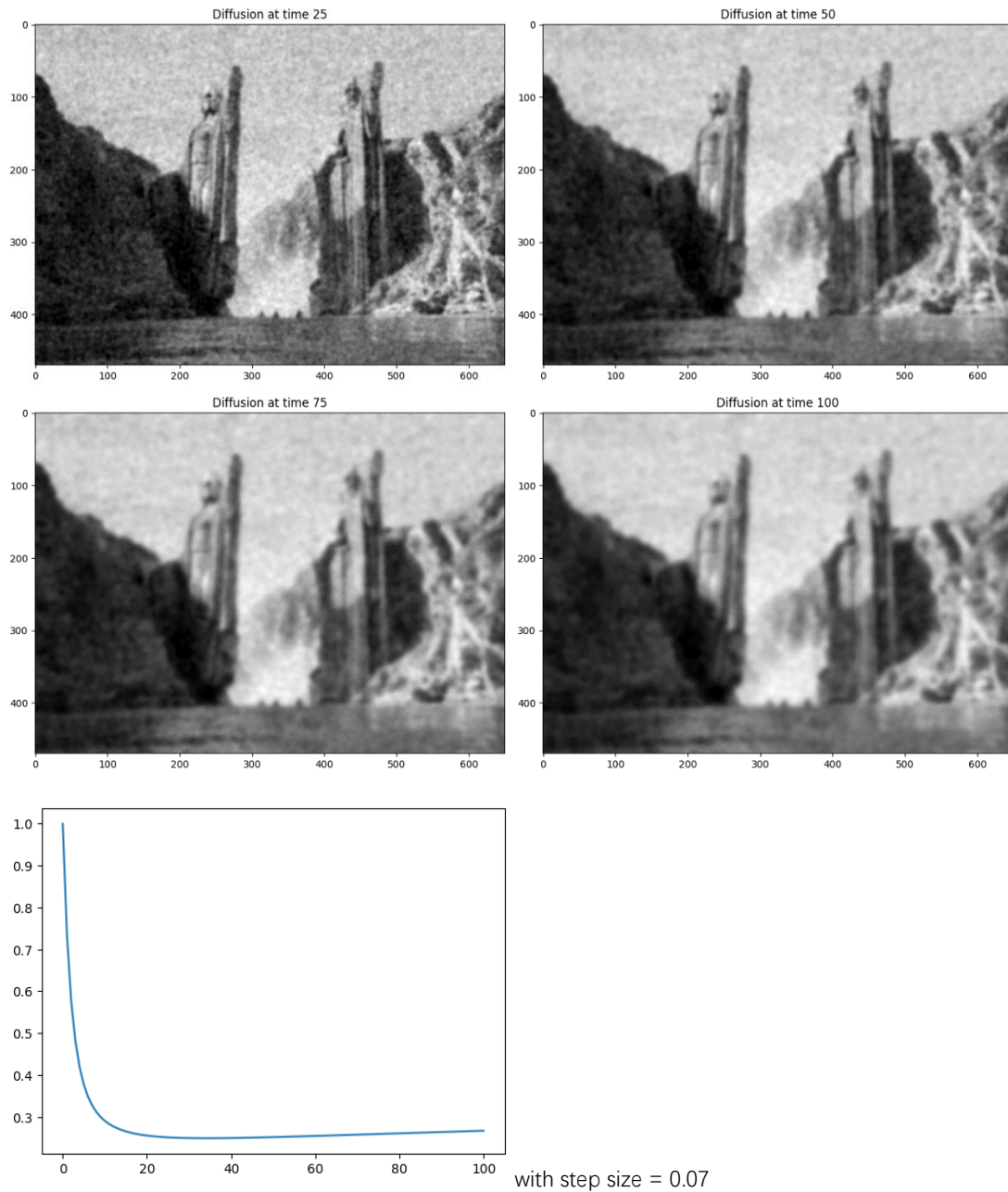
In case implementation of the Gaussian Kernel and the convolution step is required, I also did them myself.



The results from `cv2.GaussianBlur()` and my own implementation are pretty similar.

Task2: heat diffusion

Implementation detail in the code and comments



Task3: variational approach

Derivation:

Derivation of the energy to obtain the Euler-Lagrange equation: From the lecture slides we know that for the functional $E(u) = \int_{\Omega} L(u, u') dx$ we have the Euler-Lagrange Equation(1D) as the necessary condition for the extrema. The Euler-Lagrange Equation states

$$\frac{dE}{du} = \frac{\partial L}{\partial u} - \frac{d}{dx} \frac{\partial L}{\partial u'} = 0.$$

Here we have the 2D case, so I searched online for the corresponding equation for 2D. In the Generalization part, "single function of several variables with single derivative" of the website https://en.wikipedia.org/wiki/Euler%E2%80%93Lagrange_equation I found the equation.

It states that: considering a function on n variables. If Ω is some surface, then $I[f] = \int_{\Omega} L(x_1, \dots, x_n, f, f_1, \dots, f_n) dx$; $f_j = \frac{\partial f}{\partial x_j}$ is extremized only if f satisfies the partial differential equation $\frac{\partial L}{\partial f} - \sum_{j=1}^n \frac{\partial}{\partial x_j} \frac{\partial L}{\partial f_j} = 0$.

In our case, The L function is $(I(\mathbf{x}) - I_0(\mathbf{x}))^2 + \lambda \|\nabla I(\mathbf{x})\|^2$, where $\mathbf{x}$ is (x, y) . So we have two variables, x and y . f is $I(x, y)$. After rewriting, we can get

$$\begin{aligned} L &= (I(x, y) - I_0(x, y))^2 + \lambda \left(\left(\frac{\partial I}{\partial x}(x, y), \frac{\partial I}{\partial y}(x, y) \right)^T \right)^2 \\ &= (I(x, y) - I_0(x, y))^2 + \lambda \left(\frac{\partial I}{\partial x}(x, y) \right)^2 + \lambda \left(\frac{\partial I}{\partial y}(x, y) \right)^2 \\ &= L(x_1, x_2, f, f_1, f_2), \text{ where } x_1 = x, x_2 = y, f = I(x, y), f_1 = \frac{\partial I}{\partial x}(x, y) \text{ and } f_2 = \frac{\partial I}{\partial y}(x, y) \end{aligned}$$

Now let's fill in the differential equation:

$$\begin{aligned} \frac{\partial L}{\partial f} &= \frac{\partial L}{\partial I(x, y)} = 2(I(x, y) - I_0(x, y)) \\ \frac{\partial L}{\partial f_1} &= 2\lambda \frac{\partial I}{\partial x}(x, y) \\ \frac{\partial L}{\partial f_2} &= 2\lambda \frac{\partial I}{\partial y}(x, y) \\ \frac{\partial}{\partial x_1} \frac{\partial L}{\partial f_1} &= 2\lambda \frac{\partial}{\partial x} \frac{\partial I}{\partial x}(x, y) = 2\lambda \frac{\partial^2 I}{\partial x^2}(x, y) \\ \frac{\partial}{\partial x_2} \frac{\partial L}{\partial f_2} &= 2\lambda \frac{\partial}{\partial y} \frac{\partial I}{\partial y}(x, y) = 2\lambda \frac{\partial^2 I}{\partial y^2}(x, y) \end{aligned}$$

So the equation we get is:

$$2(I(x, y) - I_0(x, y)) - 2\lambda \frac{\partial^2 I}{\partial x^2}(x, y) - 2\lambda \frac{\partial^2 I}{\partial y^2}(x, y) = 0$$

Dividing both sides by 2:

$$\begin{aligned} (I(x, y) - I_0(x, y)) - \lambda \frac{\partial^2 I}{\partial x^2}(x, y) - \lambda \frac{\partial^2 I}{\partial y^2}(x, y) &= 0 \\ (I(x, y) - I_0(x, y)) - \lambda \left(\frac{\partial^2 I}{\partial x^2}(x, y) + \frac{\partial^2 I}{\partial y^2}(x, y) \right) &= 0 \end{aligned}$$

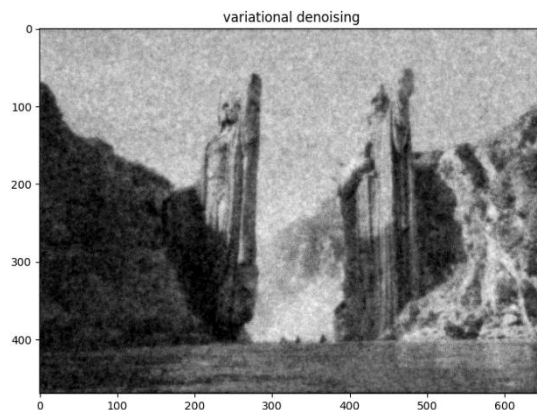
where $\frac{\partial^2 I}{\partial x^2}(x, y) + \frac{\partial^2 I}{\partial y^2}(x, y)$ is also the same as $\text{div}(\nabla I(x, y))$

After rearranging, we get:

$$\begin{aligned} (I(x, y) - I_0(x, y)) - \lambda \text{div}(\nabla I(x, y)) &= 0 \\ I_0(x, y) &= I(x, y) - \lambda \text{div}(\nabla I(x, y)) \end{aligned}$$

which is the same as equation (5) in the exercise sheet:

$$I_0 = I_u - \lambda \text{div}(\nabla I_u)$$



Task4: Comparison / Questions

- When running the gaussian convolution over and over, does the solution converge? If so, what would be the steady state? Does it depend on the boundary conditions (Neumann vs Dirichlet)?

Answer: Yes. Applying a gaussian convolution to an image is the same as using a low pass filter. Convolving an image with the gaussian filter is the same as multiplying the image's frequency space with the gaussian filter's. Convolving in the image space over and over is the same as multiplying in the frequency space over and over. The intensities of image's frequencies that multiplied with a gaussian intensity bigger than 1 goes to infinity, while the intensities of the images frequencies multiplied with a gaussian intensity smaller than 1 tends to 0. Since the gaussian filter's frequency space is another gaussian with center at zero, the intensities of the low frequency part of the image get amplified while the high frequency part get eliminated. The final image will be the low frequency part's image space clipped between 0 and 255, which is the steady state. The steady state depends on the size of the Gaussian filter than we use. The bigger its standard deviation, the lower frequencies we have in the steady state.

Yes, boundary conditions matter. E.g. we can consider the borders of Dirichlet as sharp edges when the value that we specify differs a lot from the borders of the image. Sharp edges are considered as high frequency part of the image, so it is easily removed by a low pass filter. On the contrary, Neumann considers the original image, thus the change on the borders are smaller, thus lower frequency. So the pattern on the borders tends to be remained after a low pass filter, conserving more information of the original image.

- If you apply heat diffusion infinitely, does it converge to a steady state? If so, what would be the steady state? Does it depend on the boundary conditions (Neumann vs Dirichlet)?

Answer: Yes. The steady state of the diffusion process is a constant image with the average gray value. Yes, it depends on the boundary conditions. The Dirichlet boundary condition may lead to potential artifacts while the Neumann boundary condition dictates the flow of information and results in smoother edges.

- What is the Euler Lagrange equation of an energy? What is its purpose?

Answer: The Euler Lagrange equation is a differential equation expressing the necessary condition for extrema of an energy function. Suppose $E(u) = \int_{\Omega} L(u, u') dx$ is the energy functional, we would like to find the function $u(x)$ that minimized the energy.

The Euler Lagrange equation $\frac{dE}{du} = \frac{\partial L}{\partial u} - \frac{d}{dx} \frac{\partial L}{\partial u'} = 0$ helps us find it.

- Is the solution of the Euler-Lagrange equation globally optimal? What parameter can you modify to change the level of smoothing? Explain in your own words how it works.

Answer: the solution of the Euler Lagrange equation might be a local extremum and is not guaranteed to be globally optimal, it depends on the specific problem and the

behavior of the functional that we want to minimize.

In the energy function of image denoising, we have separate cost for data similarity and smoothness. The smoothness term is a regularization term multiplied with λ . Increasing λ leads to more attention to the smoothness part and thus aggressive smoothing, but potentially at the expense of losing fine details. On the other hand, lower λ values preserve more details but may not effectively reduce noise.

- How can you describe the results? Does any of these methods give better results than the others? Explain briefly your answer.

Answer: The result images deal with the abrupt sparkling of the white noise points/intense intensity changes in the noisy image. The noise points are smoothed out. However, I don't think the results are good. For filtering and heat diffusion, with few iterations, we can still see inconsistent colors where should have a constant color. With more iterations, colors are more consistent, but the image is too blur to see the details.

From the result images, I think the three methods give similar results. The only image of variational method is similar to the best ones (e.g. iteration 8 of filtering and 25 of heat diffusion) of the other two methods. The lecture slides also say that all three methods compute the same result with appropriate parameters.

- What are the benefits and drawbacks of each method?

Answer:

Gaussian filter:

- Benefits:
 - Easy to implement, computationally efficient.
- Drawback:
 - Choice of kernel size as a hidden parameter to tune.
 - Local instead of global optimality criterion.

Heat diffusion:

- Benefits:
 - Small neighborhood structure, needs less memory compared to Gaussian filter with large standard deviation.
 - Provides a natural framework for denoising based on the diffusion of intensity values.
 - Can adjust parameters to control the level of smoothing.
- Drawback:
 - When should the diffusion be stopped needs to be found out.
 - Sensitive parameter selection. Step size as a hidden parameter to choose, need to be small for stability (which means slow convergence).

Variational approach:

- Benefits:
 - Transparent formulation with global optimality criterion.

- Many optimization methods available, solving a linear system can be much faster than computing diffusion iterations.
 - Concept of separating data and smoothness cost generalized well and is a common pattern.
 - Non-linear diffusivity (e.g. for edge preservation) is easy to incorporate.
 - Drawbacks:
 - Computational complexity can be high, solving least squares for $(mn) \times (mn)$ matrix, where m and n are image's height and width.
 - Choice of λ need to be tuned.
- Can you explain the motivations / intuitions behind each of the methods?
 - Gaussian filtering: similar to the first question of this task. Noises are high frequency, By convolving with a gaussian filter, the high frequency part of the image is filtered out and only low frequency part passes through, thus removing the noise.
 - Heat diffusion: Mimicking how heat naturally spreads to smooth out temperature variations. Here we treat regions with noise and regions without noise as high and low temperatures.
 - Variational methods: find the image that minimizes the energy that takes both data similarity and smoothness into consideration. Try to find the best image that simultaneously similar to the observed noisy image, and achieves smoothness (removes noise).